Scenario 1: Firewall Rule Efficiency Evaluation
A Linux server's firewall logs indicate that, on average, 10,000 connection attempts are made per day. Out of these, 2,500 are blocked based on predefined security rules. Calculate the percentage of connection attempts that are blocked daily. If the goal is to block at least 30% of connection attempts, determine whether the current firewall rules meet this objective.
Solution:
The percentage of blocked connection attempts is calculated as:
Blocked Percentage=(Blocked Attempts / Total Attempts)×100
Plugging in the values:
Blocked Percentage=(2,500/10,000)×100=25%
Since 25% < 30%, the current firewall rules do not meet the objective of blocking at least 30% of connection attempts.

Scenario 2: Intrusion Detection System (IDS) Performance Metrics
An IDS deployed on a Unix network generates alerts for potential security incidents. Over a month, the IDS recorded 1,200 alerts, of which 960 were true positives (actual threats) and 240 were false positives (benign activities incorrectly flagged). Calculate the following performance metrics for the IDS:
a. True Positive Rate (TPR)
b. False Positive Rate (FPR)
Solution:
a. True Positive Rate (TPR), also known as sensitivity or recall, is calculated as:
TPR=True Positives / (True Positives+False Negatives)
However, the number of false negatives is not provided. Assuming all actual threats were detected (i.e., false negatives = 0):
TPR=960/(960+0)=1 or 100%
b. False Positive Rate (FPR) is calculated as:
FPR=False Positives / (False Positives + True Negatives)
The number of true negatives is not provided. Therefore, FPR cannot be calculated with the given data. Additional information on true negatives is required to compute the FPR.

Scenario 3: Firewall Rule Optimization Analysis
A company has a firewall configured with **500 rules** for filtering inbound and outbound traffic. Each rule processes **100 packets per second** on average. The firewall has a **processing capacity of 50,000 packets per second**.

1. Calculate the total number of packets processed per second by all rules.
2. Determine whether the firewall is operating within its capacity.
3. If the company plans to add **200 more rules**, estimate the new total packet load and check whether it will exceed the firewall's capacity.

Solution:

1. **Total Packets Processed Per Second (Current Setup):**

Total Packets=Number of Rules×Packets per Rule =500×100=50,000 packets/second

2. **Check Against Capacity:**
   The firewall **capacity is 50,000 packets per second**, which means it is currently operating at **100% of its limit**.
3. **New Packet Load After Adding 200 More Rules:**

New Total Packets=(500+200)×100=700×100=70,000 packets/second
Since **70,000 > 50,000**, the firewall will be **overloaded** after adding the new rules. **A hardware upgrade or rule optimization is needed** to handle the increased load.

Scenario 4: Data Transfer and Encryption Overhead
A Unix-based security system transfers **20 GB of log data daily** to a remote server. The company decides to implement **AES-256 encryption**, which adds an overhead of **15% to the file size**.

1. Calculate the **total data size after encryption**.
2. If the network bandwidth is **500 Mbps**, calculate the **total transfer time in seconds** for the encrypted data.

Solution:

1. **Data Size After Encryption:**

Encrypted Data Size=Original Data+(Original Data×Overhead Percentage) =20+(20×0.15)=20+3=23 GB

2. **Convert Data to Bits:**

Since **1 GB = 8,000 Megabits (Mb)**, the encrypted data size in megabits is:
23×8,000=184,000 Mb

3. **Transfer Time Calculation:**

Transfer Time=Total Data in Mb / Network Speed in Mbps
=184,000 / 500=368 seconds
Thus, the encrypted data transfer will take **368 seconds (6 minutes and 8 seconds)**.

Scenario 5: Load Balancer Efficiency in Network Traffic
A security team deploys a **load balancer** to distribute network traffic across **four intrusion detection servers**. The total incoming traffic is **200 Mbps**, and the load balancer distributes traffic equally among the servers.

1. Calculate the **traffic each server receives**.
2. If each server can handle a **maximum of 60 Mbps**, determine whether they can operate without being overloaded.
3. If **one server fails**, how much traffic will each remaining server handle, and will they still operate within capacity?

Solution:

1. **Traffic Per Server:**

Traffic Per Server = Total Traffic / Number of Servers
Total Traffic =200/4=50 Mbps per server

2. **Check Against Capacity:**
   Each server can handle **60 Mbps**, and they are currently processing **50 Mbps**, so they are operating **within safe limits**.
3. **Traffic Redistribution After One Server Fails:**

New Traffic Per Server=Total Traffic / Remaining Servers =200/3=66.67 Mbps per server
Since **66.67 Mbps > 60 Mbps**, the remaining servers **cannot handle the load**, and the system will experience packet loss or require an immediate scaling solution.

Scenario 6: Log File Rotation and Storage Management
A Unix security system generates **5 log files per hour**, each of size **250 MB**. Logs are retained for **7 days**, after which they are archived.

1. Calculate the **total storage needed for 7 days**.
2. If the storage capacity is **3 TB**, determine whether it is sufficient.
3. If storage is **insufficient**, calculate how many days the system can retain logs before it runs out of space.

Solution:

1. **Daily Log Generation:**

Logs Per Day=5×24=120 logs
Daily Storage=120×250=30,000 MB=30 GB

2. **Total Storage for 7 Days:**

Total Storage=30×7=210 GB

3. **Check Against Capacity:**

Available Storage=3 TB=3,000 GB
Since **210 GB < 3,000 GB**, the storage is **sufficient**.

4. **If Storage is Limited to 500 GB:**

Max Retention Days=500/30≈16.67 days
Thus, if only **500 GB is available**, logs can be stored for about **16 days** before running out of space.

Scenario 7: Network Bandwidth Allocation for Security Applications

A Unix server runs **three security applications**, each requiring different bandwidth allocations:

- **IDS (Intrusion Detection System):** 200 Mbps
- **VPN Service:** 150 Mbps
- **Firewall Logging System:** 100 Mbps

The total **network bandwidth is 500 Mbps**.

1. Calculate the **total bandwidth required**.
2. Determine whether the network can handle the load.
3. If the firewall logging system is upgraded and now requires **50% more bandwidth**, recalculate the total bandwidth required and determine if the network remains within capacity.

Solution:

1. **Total Bandwidth Required (Before Upgrade):**

200+150+100=450 Mbps

2. **Check Against Capacity:**
   The network has **500 Mbps**, and it currently uses **450 Mbps**, so it is **operating within limits**.
3. **New Firewall Bandwidth Requirement:**

New Firewall Bandwidth=100+(100×0.50)=100+50=150 Mbps

4. **New Total Bandwidth Requirement:**

200+150+150=500 Mbps

Since the total bandwidth **matches the available capacity exactly**, the network **can handle the load, but has no margin for additional traffic**.